

Exploratory Data Analysis (EDA) Report

HR Analytics Dashboard

1. Introduction

This Exploratory Data Analysis (EDA) report analyzes employee data to understand the organization's **workforce structure, recruitment activities, employee performance, attrition, attendance, satisfaction, overtime and leave patterns.**

The analysis is presented through five Power BI dashboard pages:

1. **Executive Summary**
2. **Recruitment Analysis**
3. **Employee Performance**
4. **Attrition Analysis**
5. **Workforce Insights**

The dataset contains approximately **1,000 employees** distributed across six departments and multiple Indian states. The dashboard provides an overall view of employee demographics as well as important HR performance indicators.

2. Objectives of the Analysis

The major objectives of this EDA are:

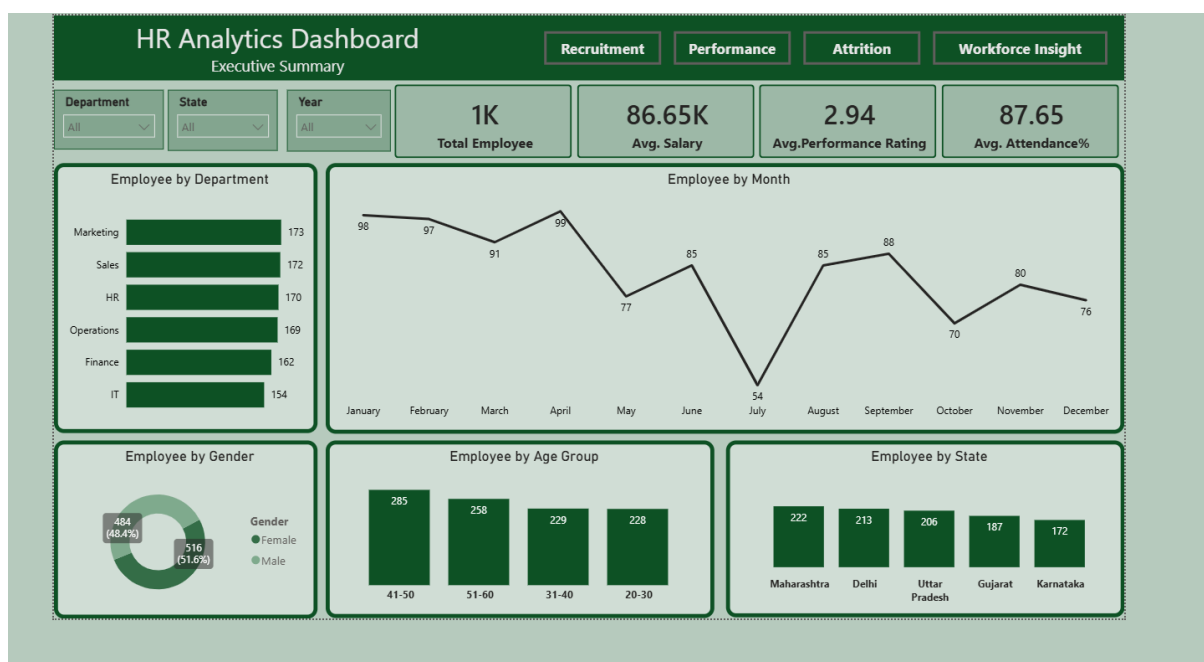
- Understand the overall employee workforce.
 - Analyze employee distribution by department, state, gender and age.
 - Examine monthly recruitment patterns.
 - Identify important recruitment sources.
 - Analyze recruitment cost and hiring efficiency.
 - Evaluate employee performance across departments.
 - Understand the relationship between training, satisfaction and performance.
 - Analyze employee attrition and identify high-risk groups.
 - Examine salary bands associated with attrition.
 - Analyze attendance, leave and overtime patterns.
 - Identify important HR trends and provide business recommendations.
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3. Dataset Overview

Metric	Dashboard Value
Total Employees	1,000

Average Salary	86.65K
Average Performance Rating	2.94
Average Attendance	87.65%
Recruitment Cost	27M
Average Hiring Days	31.18 days
Cost per Hire	27.45K
Total Employees Left	207
Attrition Rate	~21%
Average Exit Salary	90.63K
Average Experience Before Exit	4.50 years
Average Training Hours	42.93 hours
Promotion Rate	~50%
High Performers	389
Average Leave Days	15.11 days
Overtime	~51%
Average Satisfaction Score	5.62

Page 1 – Executive Summary Analysis



Employee Distribution by Department

The organization has six major departments:

Department Employees

Marketing 173

Sales	172
HR	170
Operations	169
Finance	162
IT	154

EDA Insight

Marketing has the highest number of employees with **173**, followed closely by Sales with **172**.

IT has the smallest workforce with **154 employees**.

The distribution is relatively balanced, with no single department dominating the workforce. This indicates a diversified organizational structure.

Employee Hiring by Month

The monthly employee distribution shows the following pattern:

Month	Employees/Hires
January	98
February	97
March	91
April	99
May	77
June	85
July	54
August	85
September	88
October	70
November	80
December	76

EDA Insight

April has the highest hiring activity with 99 employees, while July has the lowest with only 54 employees.

The sharp decline during July could indicate seasonal recruitment patterns, hiring-budget cycles or changes in workforce demand.

Employee Gender Analysis

The dashboard shows:

- **Male:** 516 employees — approximately **51.6%**
- **Female:** 484 employees — approximately **48.4%**

EDA Insight

The gender distribution is relatively balanced, with a difference of only around **3.2 percentage points**.

This indicates a comparatively balanced workforce from a gender perspective.

Employee Age Group Analysis

The employee distribution by age group is:

Age Group Employees

41–50	285
51–60	258
31–40	229
20–30	228

EDA Insight

The largest employee group is **41–50 years**, with 285 employees.

Employees aged **41–60 together represent more than half of the workforce**, indicating a relatively experienced workforce.

The organization may therefore need to focus on:

- Succession planning
- Leadership development
- Knowledge transfer
- Retirement planning
- Career progression

Employee Distribution by State

The major states shown in the dashboard are:

State	Employees
Maharashtra	222

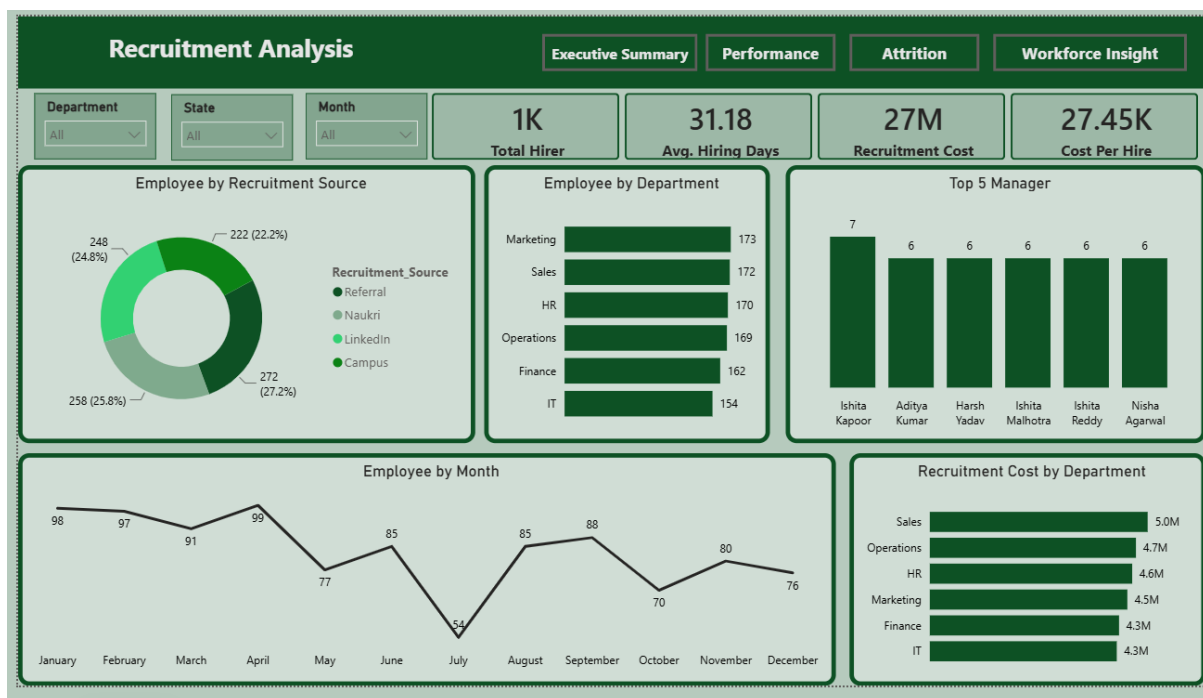
Delhi **213**
 Uttar Pradesh **206**
 Gujarat **187**
 Karnataka **172**

EDA Insight

Maharashtra has the highest workforce representation, while Karnataka has the lowest among the displayed states.

The organization has a geographically distributed workforce, with Maharashtra, Delhi and Uttar Pradesh accounting for a large share of employees.

Page 2 – Recruitment Analysis



Recruitment Sources

The recruitment source distribution is:

Recruitment Source	Employees	Approx. Share
Referral	272	27.2%
Naukri	258	25.8%
LinkedIn	248	24.8%
Campus	222	22.2%

EDA Insight

Employee referrals are the largest recruitment source, contributing approximately 27.2% of hires.

The distribution is fairly balanced, which means the organization does not depend heavily on a single recruitment channel.

However, referrals, Naukri and LinkedIn together account for approximately **78% of recruitment**, making these channels particularly important.

Recruitment Efficiency

The dashboard reports:

- **1K total hires**
- **31.18 average hiring days**
- **27M recruitment cost**
- **27.45K cost per hire**

EDA Insight

An average hiring time of approximately **31 days** indicates that the recruitment process takes around one month per hire.

The organization should monitor hiring time by:

- Department
- Recruitment source
- Job role
- Location
- Experience level

This can identify channels that provide faster and more cost-effective hiring.

11. Recruitment Cost by Department

Approximate recruitment costs are:

Department Recruitment Cost

Sales	5.0M
Operations	4.7M
HR	4.6M
Marketing	4.5M
Finance	4.3M
IT	4.3M

EDA Insight

Sales has the highest recruitment cost at approximately **5.0M**, while Finance and IT have the lowest at around **4.3M**.

The difference is relatively moderate and appears to be related partly to workforce size.

For a deeper analysis, recruitment cost should be normalized using:

Cost per Hire by Department = Recruitment Cost / Number of Hires

Top Managers

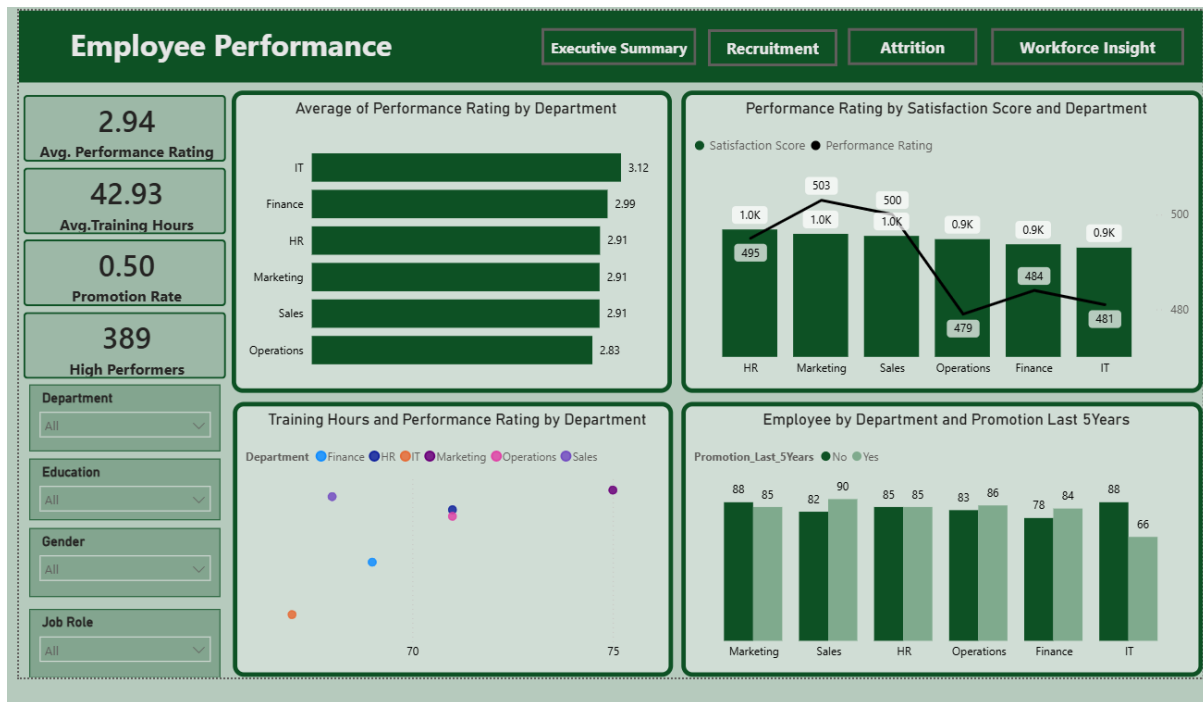
The dashboard identifies six managers with approximately **6 employees each**, while one manager has **7 employees**.

EDA Insight

Managerial workload appears relatively balanced based on team size.

However, team size alone cannot measure managerial effectiveness. Future analysis should compare manager-wise:

- Employee performance
- Attrition
- Satisfaction
- Attendance
- Promotion
- Overtime



Performance Rating by Department

Department Average Performance Rating

IT	3.12
Finance	2.99
HR	2.91
Marketing	2.91
Sales	2.91
Operations	2.83

EDA Insight

IT has the highest average performance rating at 3.12, while Operations has the lowest at 2.83.

This indicates a noticeable difference in performance across departments.

Operations should be examined further to determine whether lower performance is associated with:

- Training gaps
- Workload
- Job role
- Satisfaction
- Management
- Attendance
- Overtime

Training and Performance

The overall average training time is approximately **42.93 hours**.

The scatter visualization compares training hours with performance rating across departments.

EDA Insight

Training is an important factor to investigate because departments with higher training exposure may demonstrate stronger performance.

However, the visualization alone does not prove causation. A statistical correlation analysis should be performed before concluding that additional training directly increases performance.

Satisfaction and Performance

The dashboard compares satisfaction scores with performance ratings.

Approximate performance-related employee counts are shown around:

- HR: 495
- Marketing: 503
- Sales: 500
- Operations: 479
- Finance: 484
- IT: 481

EDA Insight

Marketing appears to have one of the strongest performance outcomes, while Operations shows comparatively lower performance.

Because satisfaction scores are also available, HR management should investigate whether lower satisfaction is associated with lower performance.

A useful future metric would be:

Average Performance Rating by Satisfaction Band

Promotion Analysis

The dashboard compares employees who received and did not receive a promotion during the last five years.

EDA Insight

Promotion patterns vary by department.

For example:

- Marketing: 88 vs 85
- Sales: 82 vs 90
- HR: 85 vs 85
- Operations: 83 vs 86
- Finance: 78 vs 84
- IT: 88 vs 66

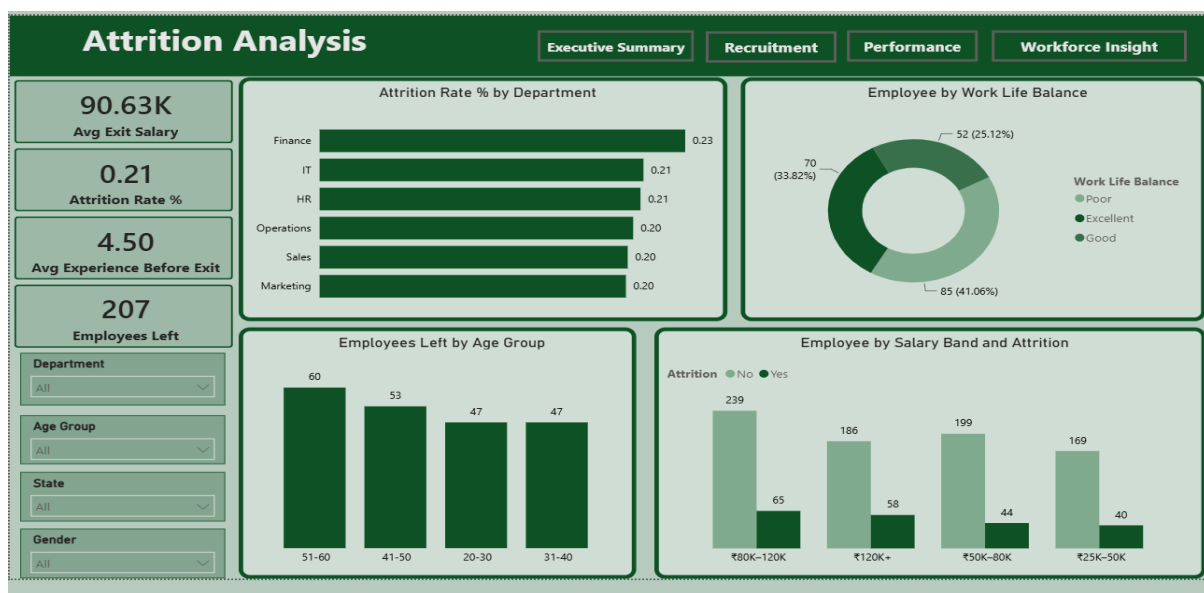
The IT department shows a notable difference between employees without and with promotion.

Recommendation

HR should investigate whether promotion opportunities are fairly distributed based on:

- Performance
- Experience
- Job role
- Training
- Tenure
- Department

Page 4 – Attrition Analysis



Overall Attrition

The dashboard reports:

- **207 employees left**
- Attrition rate: approximately **21%**
- Average exit salary: **90.63K**
- Average experience before exit: **4.50 years**

EDA Insight

An attrition rate of approximately **21% is significant** and should be closely monitored.

The organization should identify the main reasons employees leave, particularly among experienced and high-performing employees.

Attrition Rate by Department

The approximate attrition rates shown are:

Department Attrition Rate

Finance	23%
IT	21%
HR	21%
Operations	20%
Sales	20%
Marketing	20%

EDA Insight

Finance has the highest attrition rate at approximately 23%, while Marketing has one of the lowest at approximately 20%.

Although the differences appear small, Finance should be investigated further to identify specific causes of employee exits.

Employee Attrition by Age Group

Among employees who left:

Age Group Employees Left

51–60	60
41–50	53
20–30	47
31–40	47

EDA Insight

The **51–60 age group has the highest number of employees leaving**, with 60 exits.

This is important because experienced employees may carry significant organizational knowledge.

The company should consider:

- Retention programs
- Succession planning
- Flexible work options
- Career development
- Knowledge-transfer programs

Attrition by Salary Band

Employees leaving by salary band:

Salary Band	Employees Left
₹80K–₹120K	65
₹120K+	58
₹50K–₹80K	44
₹25K–₹50K	40

EDA Insight

The highest number of exits occurs among employees earning **₹80K–₹120K**, followed by employees earning **₹120K+**.

This is an important finding because attrition is not concentrated only among lower-paid employees.

Higher-paid employees may be leaving due to:

- Better external opportunities
- Career growth
- Work-life balance
- Leadership
- Job satisfaction
- Promotion opportunities

Work-Life Balance of Employees Who Left

Among employees who left:

- **Good:** 85 — approximately **41.06%**

- **Poor:** 70 — approximately **33.82%**
- **Excellent:** 52 — approximately **25.12%**

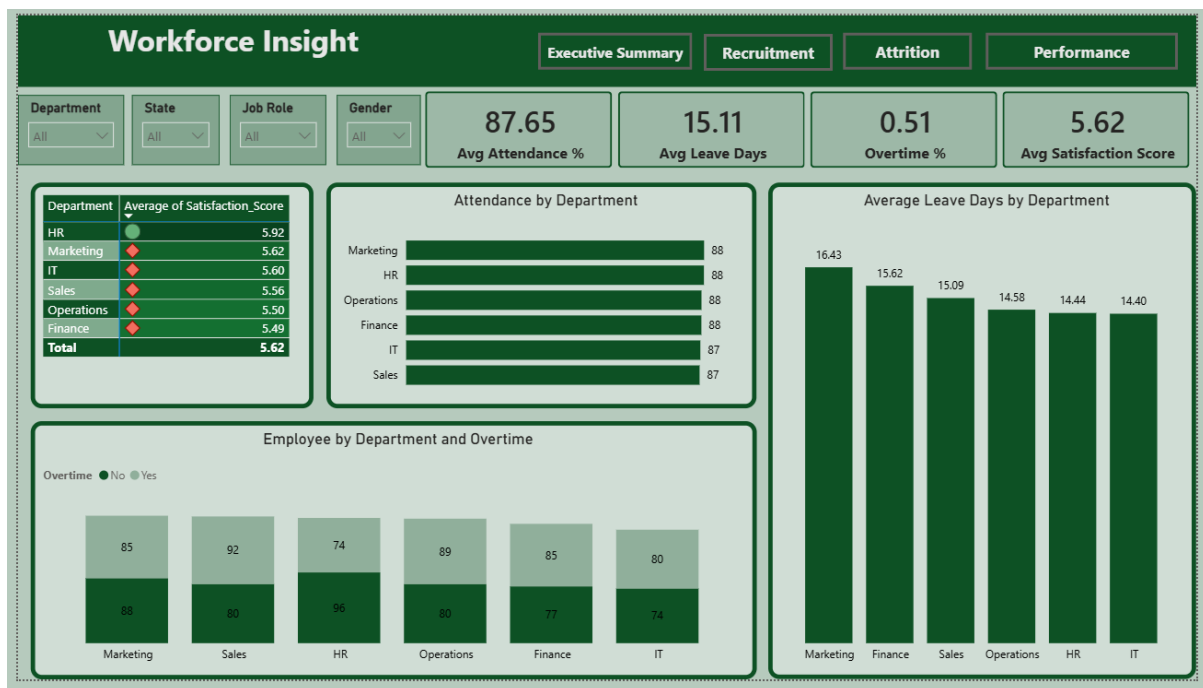
EDA Insight

A significant proportion of employees who left had **Good or Excellent work-life balance**.

This indicates that work-life balance alone may not explain attrition.

Other factors such as compensation, career growth, job role, management and promotion should therefore be investigated.

Page 5 – Workforce Insights



Employee Satisfaction

Average satisfaction score:

5.62

Department-wise satisfaction:

Department Avg. Satisfaction

HR 5.92

Marketing 5.62

IT 5.60

Sales	5.56
Operations	5.50
Finance	5.49

EDA Insight

HR has the highest satisfaction score at **5.92**, while Finance has the lowest at **5.49**.

Finance therefore represents an important department for further employee-engagement analysis, especially because it also has the highest attrition rate.

Attendance by Department

Average attendance is approximately **87.65%**.

Department-level attendance is approximately:

- Marketing: 88%
- HR: 88%
- Operations: 88%
- Finance: 88%
- IT: 87%
- Sales: 87%

EDA Insight

Attendance levels are relatively consistent across departments.

The small variation suggests that attendance is not dramatically different between departments.

However, attendance should be compared with performance and attrition to identify hidden relationships.

Leave Analysis

Average leave usage is approximately **15.11 days**.

Department-wise average leave:

Department	Avg. Leave Days
Marketing	16.43

Finance	15.62
Sales	15.09
Operations	14.58
HR	14.44
IT	14.40

EDA Insight

Marketing has the highest average leave usage at **16.43 days**, while IT has the lowest at **14.40 days**.

Higher leave usage should not automatically be considered negative because employees may be using legitimate annual leave.

The important analysis is whether high leave is associated with:

- Lower performance
- Higher satisfaction
- Lower burnout
- Higher or lower attrition

Overtime Analysis

The workforce dashboard shows overtime distribution by department.

Approximate overtime/non-overtime counts indicate that overtime is common across departments.

For example:

- Marketing: 88 No / 85 Yes
- Sales: 80 No / 92 Yes
- HR: 96 No / 74 Yes
- Operations: 80 No / 89 Yes
- Finance: 77 No / 85 Yes
- IT: 74 No / 80 Yes

EDA Insight

Sales has one of the highest numbers of employees working overtime.

IT and Finance also show relatively high overtime levels.

This is important because excessive overtime can potentially affect:

- Employee satisfaction
- Work-life balance
- Productivity
- Performance
- Attrition

Data Quality and Validation Checks

Before using the dashboard for management decisions, the following checks should be performed:

1. Attrition Rate Validation

The dashboard shows:

207 employees left / 1,000 employees = 20.7%

Therefore, the displayed **0.21** is approximately **21%**.

The KPI should ideally be formatted as **20.7% or 21%**, rather than simply 0.21.

2. Promotion Rate Validation

The dashboard displays a promotion rate of **0.50**.

For management reporting, it would be clearer to display this as:

50%

provided the underlying calculation confirms this interpretation.

3. Recruitment Cost Validation

The dashboard displays:

- Recruitment Cost = 27M
- Cost per Hire = 27.45K
- Total Hires = approximately 1,000

The exact measures should be validated because 27M divided by 1,000 is approximately 27K, while the dashboard displays 27.45K.

4. Age Group Definitions

Age bands should be standardized:

- 20–30
- 31–40

- 41–50
- 51–60

The organization should verify that boundary ages are consistently classified.

5. Department Totals

The department counts add up to approximately **1,000 employees**, which is consistent with the overall employee KPI.

Business Recommendations

1. Reduce Finance Attrition

Finance has the highest attrition rate and lowest satisfaction score. Conduct employee surveys and exit interviews to identify specific reasons for employee turnover.

2. Focus on Experienced Employees

Employees aged 41–60 represent a large portion of the workforce and also account for many exits. Introduce retention and succession-planning programs.

3. Monitor High-Salary Attrition

Employees in higher salary bands are leaving in significant numbers. Compensation alone may not be sufficient to retain them, so career growth and leadership opportunities should also be evaluated.

4. Improve Operations Performance

Operations has the lowest average performance rating. Review training, workload, employee engagement and managerial support.

5. Optimize Recruitment Sources

Referrals are currently the strongest recruitment source. HR should compare each source based on:

Hiring Cost + Hiring Time + Employee Performance + Retention

rather than simply measuring the number of hires.

6. Monitor Overtime

Departments with high overtime participation should be reviewed for workload imbalance and employee burnout risks.

7. Strengthen Employee Engagement

Finance and Operations have relatively weaker satisfaction/performance indicators. Targeted employee engagement programs can help improve retention and productivity.

8. Use Data-Driven Promotion Planning

Promotion decisions should be evaluated against performance, experience, training and tenure to ensure that high-performing employees have clear career-development opportunities.

Final EDA Summary

Area	Key Finding
Workforce	1,000 employees
Largest Department	Marketing – 173
Smallest Department	IT – 154
Gender	51.6% Male / 48.4% Female
Largest Age Group	41–50 – 285 employees
Highest Hiring Month	April – 99 hires
Lowest Hiring Month	July – 54 hires
Best Recruitment Source	Referral – 272 hires
Recruitment Cost	27M
Avg. Hiring Time	31.18 days
Best Performance Department	IT – 3.12
Lowest Performance Department	Operations – 2.83
Employees Left	207
Overall Attrition	~21%
Highest Attrition	Finance – ~23%
Highest Exit Age Group	51–60 – 60 employees
Highest Exit Salary Band	₹80K–₹120K – 65 employees
Highest Satisfaction	HR – 5.92
Lowest Satisfaction	Finance – 5.49
Avg. Attendance	87.65%
Avg. Leave	15.11 days
Avg. Overtime	~51%